geographic location and at what zoom level the image shows. The `d3.geo.tile()` function (the library to access this function is available at https://github.com/d3/d3-plugins/tree/master/geo/tile) parses that filename and directory structure for us so that we can use these tiles in our map. First, though, we have to calibrate the scale and translate of our projection as well as our zoom behavior.

Listing 7.19 A tile map

```
var width = 500,
    height = 500;

d3.select("svg").append("g").attr("id", "tiles");

var tile = d3.geo.tile()
    .size([width, height]);

var projection = d3.geo.mercator()
    .scale(120)
    .translate([width / 2, height / 2]);

var center = projection([12, 42]);

var path = d3.geo.path()
    .projection(projection);

var zoom = d3.behavior.zoom()
    .scale(projection.scale() * 2 * Math.PI)
    .translate([width - center[0], height - center[1]])
    .on("zoom", redraw);

d3.select("svg").call(zoom);
    redraw();

function redraw() {
   var tiles = tile
      .scale(zoom.scale())
      .translate(zoom.translate())();

   var image = d3.select("#tiles")
      .attr("transform",
      "scale(" + tiles.scale + ") translate(" + tiles.translate + ")")
     .selectAll("image")
     .data(tiles, function(d) { return d; });

   image.exit()
      .remove();

   image.enter().append("image")
      .attr("xlink:href",
          function(d) { return "http://" +
          ["a", "b", "c", "d"][Math.random() * 4 | 0] +
          ".tiles.mapbox.com/v3/examples.map-zgrqqx0w/" +
          d[2] + "/" + d[0] + "/" + d[1] + ".png"; })
      .attr("width", 1)
      .attr("height", 1)
      .attr("x", function(d) { return d[0]; })
      .attr("y", function(d) { return d[1]; });
};
```

A group to keep our tiles behind any other drawn features

The function we use to create your tiles

The dataset we use to create the images

Generates proper transform settings based on the current zoom

Binds the tiles data to svg:image elements

Removes any that are offscreen

Appends new ones

Path to the tiles is generated by tile.js for services like Mapbox